PRACTICAL 3

Information borrowing: case of borrowing on historical control

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In this practical you are going to replicate the analysis of the motivating example of the Crohn's disease covered in the lectures and conduct some sensitivity analysis.

The study is planned to have 2:1 allocation in favour of the experimental arm. The primary endpoint is Crohn's Disease Activity Index (CDAI) change from baseline at week 6. The improvement in the patient implies the negative change from baseline. This endpoint is assumed to be normally distributed.

There are six RCTs that have used the same control arm. The summary data is as follows:

	Study	Sample Size	Mean	Standard Error
1	Gastr06	74	-51	10.23
2	AIMed07	166	-49	6.83
3	NEJM07	328	-36	4.86
4	Gastr01a	20	-47	19.68
5	APhTh04	25	-90	17.60
6	Gastr01b	58	-54	11.55

We would like to use these data to construct an information prior for the control group. The vague prior is going to be used for the experimental arm. The decision criteria is based on the posterior probability criterion. If the posterior probability of the positive treatment effect is greater than 97.5% then the drug is effective.

1. Using the code "Practical_3_Template.R" replicate the analysis of using the MAP prior with the weight of the robustification component being equal to 20%. What is the effect sample size of the MAP prior and the robustified MAP prior?

2. Assume that at the end of the trial 39 patients were assigned to the experimental arm and the mean response was -100. On the placebo arm, 19 patients were assigned and the mean response was -60. Is this mean response in the placebo group aligned with the historical data used in the rMAP? Why?

3. Using the code provided, complete the table below

Evidence source	Treatment difference: Active-Placebo	Pr(Treat diff) <0	Active effect	Placebo effect
Primary analysis - ***posterior***				
Current study only				
Historical MAP ***prior*** for placebo effect				
Robust MAP ***prior*** for placebo effect				

4. Assume now that the one is less confident in the relevance of the historical data in the current study and would like to use the robustification weight of 50%. What is the new ESS of the rMAP? How would this change the table above? Would it change the conclusion?

Evidence source	**Treatment difference: Active-Placebo**	**Pr(Treat diff) <0**	**Active effect**	**Placebo effect**
Primary analysis - ***posterior***				
Historical MAP ***prior*** for placebo effect				
Robust MAP ***prior*** for placebo effect				

5. Assume now that the actual summary statistics in the current study that were observed are -170 in the experimental and -130 on the control. We again use the robustification weight of 20%. How does this change the table from point 3? Does it change the conclusion of the trial?

Evidence source	**Treatment difference: Active-Placebo**	**Pr(Treat diff) <0**	**Active effect**	**Placebo effect**
Primary analysis - ***posterior***				
Current study only				
Historical MAP ***prior*** for placebo effect				
Robust MAP ***prior*** for placebo effect				

6. Assume now that you are at the design stage and you would like to understand the implication of borrowing information on the type I error in case there is a drift between the historical and current data information. The pre-specified criterion for claiming the success is as follows:

$$P(\theta_{active} - \theta_{control} < 0) > 0.975$$

Compare two design options:

1. 40:20 sample size and no borrowing on the control arm
2. 40:20 sample size and borrowing on the control arm as per rMAP above with the robustification weight of 0.2

Using the code "Practical_3_Template.R" evaluate the operating characteristics of these designs in terms of the type I error for the different values of the drift, i.e. the true mean response on the control arm is not only -50 but can taken various values from -120 to -40. Plot the type I error (using the posterior probability criterion above) as a function of the drift. What is the type I error at the case of no drift? What is the minimum and maximum type I error? Is it concerning?

7. How do the operating characteristics evaluated above change if one considers now the rMAP approach with the robustification weight of 0.5 (instead of 0.2). How does the type I error change? Which design would you prefer? What are the downsides (if any) using more conservative design?